

CSE563 — Assignment 6

Unit VI: Algorithmic and Optimization Approaches

Maps to CO6 (discuss different optimization approaches to solve mathematical problems). **Marks: 50.** **Attempt all questions. Show DP tables in full and state the traceback.**

Part A — Short Answer and Definitions (10 marks, 2 each)

Q1. *[2 marks]* State the two properties a problem must have for a greedy algorithm to be correct. Give one problem where greedy works and one where it provably fails.

Q2. *[2 marks]* Explain the difference between dynamic programming and divide-and-conquer. Which property decides between them?

Q3. *[2 marks]* Define backtracking and branch and bound, and state the difference between them.

Q4. *[2 marks]* Name the three genetic operators and state the role of each in balancing exploration and exploitation.

Q5. *[2 marks]* State the No Free Lunch theorem and explain its practical implication for choosing an optimisation algorithm.

Part B — Exact Algorithm Design (26 marks)

Q6. [6 marks] Nine activities with (start, finish) times: (1, 4), (3, 5), (0, 6), (5, 7), (3, 8), (5, 9), (6, 10), (8, 11), (12, 16).

(a) State the greedy rule and justify it in one sentence. (b) Apply it, showing the selected set. (c) Show that the rule “pick the activity with the shortest duration” can be sub-optimal by constructing a small counterexample.

Q7. [7 marks] Solve the 0/1 knapsack instance with capacity $W = 8$ and items $I_1 = (w=2, v=3)$, $I_2 = (3, 4)$, $I_3 = (4, 5)$, $I_4 = (5, 8)$. (a) Build the full DP table $dp[i][c]$ for $i = 0..4$, $c = 0..8$. (b) State the optimal value. (c) Trace back to identify which items are chosen and verify the weight fits. (d) State the time complexity and explain why it is called *pseudo-polynomial*.

Q8. [6 marks] For $X = \text{AGGTAB}$ and $Y = \text{GXTXAYB}$: (a) Build the LCS dynamic-programming table in full. (b) State the LCS length. (c) Trace back to recover the actual subsequence. (d) State the time and space complexity, and name one real application.

Q9. [7 marks] (a) Explain the backtracking approach to the N -Queens problem and state the pruning conditions used. (b) Trace the search for $N = 4$ from an empty board, listing each partial placement and whether it is extended or pruned. (c) Give both solutions for $N = 4$. (d) Compare the number of nodes explored with brute force over all 4^4 placements.

Part C — Metaheuristics (14 marks)

Q10. [8 marks] A genetic algorithm maximises $f(x) = x^2$ for $x \in [0, 31]$ using 5-bit binary encoding. The initial population is $P_1 = 01100$, $P_2 = 11001$, $P_3 = 00101$, $P_4 = 10010$. (a) Decode each chromosome and compute its fitness. (b) Compute the selection probability and expected number of copies under roulette-wheel selection. (c) Perform single-point crossover between (P_2, P_4) at position 2 and between (P_2, P_1) at position 3, and decode the children. (d) Apply a mutation flipping bit 1 (the most significant bit) of the first child, and give its new fitness. (e) Comment on whether the generation improved the population.

Q11. [6 marks] (a) Write the simulated-annealing acceptance rule and compute the acceptance probability of a move with $\Delta = +5$ at $T = 10$, $T = 2$ and $T = 0.5$ (use $e^{-0.5} = 0.6065$, $e^{-2.5} = 0.0821$, $e^{-10} = 4.54 \times 10^{-5}$). (b) Explain, using these numbers, how the algorithm shifts from exploration to exploitation. (c) Write the PSO velocity update and perform one update for a particle with $x = 3$, $v = -1$, personal best $p = 5$, global best $g = 8$, $w = 0.7$, $c_1 = c_2 = 2$, $r_1 = 0.4$, $r_2 = 0.6$.